

Enhanced FUR: Experimental Setup

Goal

Full FUR unlearns every CoT step, which is expensive. Enhanced FUR asks whether we can unlearn only a selected top-1 step and still recover many faithful cases found by Full FUR.

Compared methods

- ▶ **Full FUR**: unlearn all segmented CoT steps.
- ▶ **Last-step top1**: unlearn only the final step.
- ▶ **Random top1**: unlearn one random step.

Run configuration

Model	Llama-3.2-3B-Instruct
Dataset	OpenBookQA
CoT cache	100 generated CoTs
Unlearning targets	80 instances
Specificity split	20 held-out samples
Method	NPO-KL, stepwise, sentenceize
Epochs / LR	2 / 3e-05
Parallelism	3 shards on A100 80GB

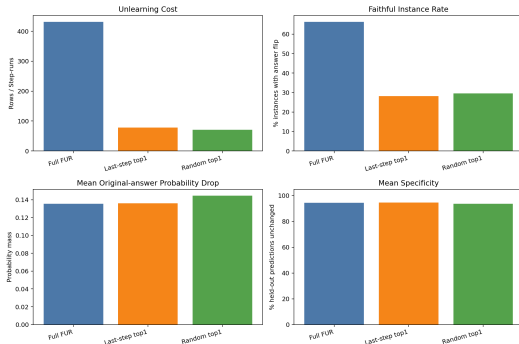
Note. Llama-8B weights were gated at download time, so the complete run uses the locally available Llama-3B model.

Overall Results: Full FUR vs. Selective Top-1

Summary metrics

Method	Rows	Faithful inst.	Specificity
Full FUR	431	53/80 = 66.25%	94.51%
Last top1	78	22/78 = 28.21%	94.74%
Random top1	71	21/71 = 29.58%	93.73%

Full FUR finds many faithful instances, while top-1 variants use far fewer unlearning runs. All methods show high efficacy, around 89–90%, and high specificity, around 94%.

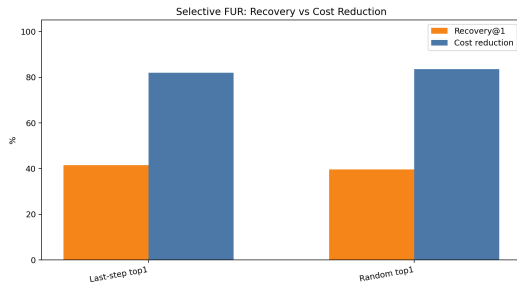


Selective FUR: Recovery–Cost Tradeoff

Recovery against Full FUR

Metric	Last top1	Random top1
Cost reduction	81.90%	83.53%
Faithful cases found	22/53	21/53
Recovery@1	41.51%	39.62%
Step-Hit@1	17.21%	18.85%
Selected-step precision	26.92%	32.39%

Top-1 selective FUR is much cheaper, but the naive selectors miss many faithful steps.
Last-step is not clearly better than random, suggesting that a real verifier/ranker is needed.

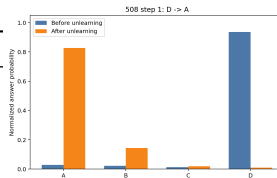


Strong Positive Cases: Probability Transfer

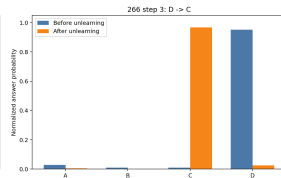
Largest Full-FUR answer shifts

ID	Step	Change	Drop	Efficacy
508	1	$D \rightarrow A$	0.927	91.56%
266	3	$D \rightarrow C$	0.926	96.09%
1189	1	$B \rightarrow A$	0.925	92.04%
1577	7	$B \rightarrow D$	0.900	95.81%
1577	11	$B \rightarrow A$	0.894	97.16%

These examples show the clearest causal effect: after one reasoning step is unlearned, the original answer probability almost vanishes and another option becomes dominant.



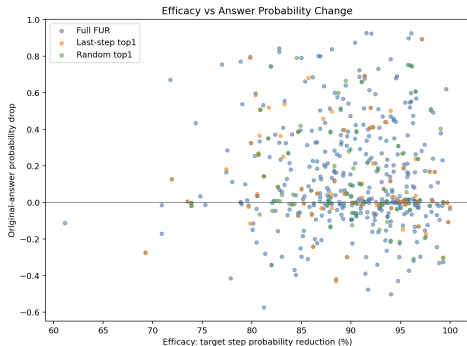
ID 508, step 1



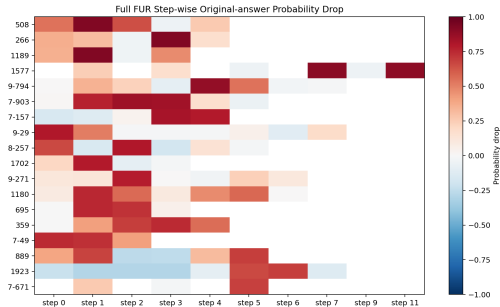
ID 266, step 3

Global Pattern and Final Takeaways

Efficacy vs. answer probability shift



Full-FUR step salience



- ▶ High efficacy does not guarantee an answer flip; probability-shift metrics are necessary.
- ▶ Full FUR is the best upper bound, but it is expensive: 431 step-runs for 80 instances.
- ▶ Top-1 selective FUR saves more than 80% of the cost, but recovers only about 40% of faithful instances.
- ▶ Next step: replace naive top-1 selection with deletion-based or verifier-guided ranking.